NewsBot Intelligence System

Technical Report | Benjamin LaCount II | ITAI 2373 | October 17, 2026

https://github.com/MarineMaaan/ITAI2373-NewsBot-Midterm

Purpose and scope

NewsBot is a CPU-based Google Colab prototype for assisted news routing and media monitoring. It combines eight NLP modules into one reusable analysis function and an interactive Gradio interface. The student works solo; no team name, team number or video is used. The notebook requires a user to download and upload the labeled BBC training CSV. Dataset approval was confirmed by the student. Raw dataset files are excluded from the public repository.

The intended operator is an editor or analyst who needs topic routing, entity leads and tone summaries. Benjamin's studio review-monitoring use case motivates future adaptation, but news-topic performance does not establish game-review performance.

Dataset and evaluation design

The retained dataset contains 1438 substantial articles after removing 52 canonical duplicates and 0 rows with missing values. It uses BBC's business, entertainment, politics, sport and tech labels. Health is outside the trained label space. There are no reliable publication timestamps in the supplied CSV.

Stratified partition sizes are 1006 training, 216 validation and 216 test articles. Three-fold training CV optimizes n-grams, feature selection and model settings. TF-IDF vocabulary and chi-squared selection are fitted within folds. Validation determines a review-margin threshold; test data does not select the classifier. Exact duplicates are removed before splitting, although near duplicates may remain.

Model	CV-selected?	Test accuracy	Test macro-F1
Naive Bayes	No	0.9722	0.9705
Logistic regression	No	0.9769	0.9757
Linear SVM	Yes	0.9815	0.9808

Main result

Training CV selected Linear SVM. Its held-out accuracy is 0.9815 and macro-F1 is 0.9808. The notebook includes per-class precision, recall, F1, support, a confusion matrix and an error table. These scores measure topic classification on this split of BBC data. They do not measure sentiment quality, entity accuracy or readiness for a live editorial workflow.

Integrated NLP and bonus evaluation

Module	Implementation and interpretation
1. Application	Editorial routing, human review and a future studio review-monitoring use case.
2. Preprocessing	Markup/Unicode/URL cleanup; tokenization, stop words and lemmatization; raw text retained.
3. TF-IDF	Fold-safe unigram/bigram search, chi-squared selection, category terms and model weights.
4. POS	Per-article POS proportions, category heatmap and grammatical bigrams.
5. Syntax	Dependency features and heuristic agent/action/patient relations, including passive voice and negation.
6. Tone	VADER sentiment and transparent four-emotion cue lexicons; no gold tone labels.
7. Classification	Naive Bayes, logistic regression and linear SVM with CV and reserved-test evaluation.
8. NER	PERSON, ORG, GPE, DATE, MONEY; frequencies, category patterns and co-occurrence map.

Bonus experiments

Interactive dashboard: Gradio accepts a new English article and returns category, tone, entities, syntax, custom entities and a review flag. Two original articles demonstrate the complete flow. The validation margin is not a calibrated probability or an out-of-domain detector.

Advanced analysis: 25 balanced original English/Spanish pairs compare English input, direct Spanish input and a fixed glossary aid. The passages intentionally contain domain keywords, making this an optimistic controlled language-sensitivity probe rather than a general multilingual benchmark.

Custom model: a small spaCy NER model learns NEWS_OUTLET and TECH_PRODUCT using 96 synthetic training, 12 development and 12 test sentences. Name pairs and sentence frames are disjoint. Development F1 selects the epoch; exact-span test precision is 0.625, recall 0.625, and F1 0.625. The no-custom-label baseline has F1 0.000. Real article performance remains unmeasured.

Findings, limits and reproducibility

Research extension

The project-specific hypothesis is that adding grammar features to TF-IDF improves topic classification. A fixed logistic regression baseline and an otherwise matched hybrid model are evaluated on the same test articles. Baseline macro-F1 is 0.9706; hybrid macro-F1 is 0.9292. The 95% stratified paired-bootstrap interval for hybrid minus baseline is [-0.0740, -0.0141] over 500 resamples.

An entity-word masking stress test produces baseline macro-F1 0.9567 and hybrid macro-F1 0.9248. Original syntax is retained in this lexical stress test. The approach is an original project experiment, not a claim of academic novelty. One split and a bootstrap conditional on fitted models cannot establish a universal benefit.

Business recommendations and limitations

Use NewsBot for assisted sorting, with a human examining uncertain cases and checking extracted entities in the source. The term and POS charts can guide exploration; they do not establish causal differences between categories. VADER measures lexical tone, and the emotion lexicons are unvalidated proxies. Co-occurrence edges are leads, not verified relationships. Lowercased BBC text weakens pretrained NER.

For studio reviews, create licensed review data and labels reflecting bugs, performance, gameplay and value. Evaluate across games, sources and time. Annotate aspect sentiment and domain entities separately. Measure operator time and error costs in a pilot before claiming productivity gains. Repeat experiments with independently annotated, representative material.

Reproduction and evidence

Open NewsBot_Intelligence_System.ipynb in Colab, run setup, and upload BBC News Train.csv when prompted. Run the remaining cells in order. The notebook embeds reusable source and exports result tables, figures and JSON. Local validation uses the same notebook in a fresh CPU kernel with an explicitly supplied CSV path. This is local execution evidence; a signed-in Colab browser session is not claimed.

Machine-readable outputs record the dataset SHA-256, category counts, split manifest, library versions and runtime. The separately supplied reflective journal is two pages and records the student's preparation, learning, studio use case and AI assistance. The four non-video bonus categories are implemented; points and the stated extra-credit cap remain the instructor's decision.

Sources and contribution

BBC News Classification: kaggle.com/competitions/learn-ai-bbc/data. Technical references: scikit-learn.org/stable/common_pitfalls.html; spacy.io/usage/linguistic-features; spacy.io/usage/training; nltk.org/api/nltk.sentiment.vader.html; gradio.app/docs/gradio/interface. VADER: Hutto and Gilbert (2014), Proceedings of ICWSM, doi:10.1609/icwsm.v8i1.14550. Full clickable references are in the notebook and README.

Benjamin LaCount II is the sole student contributor. AI assisted execution, code, debugging, evaluation and documentation. The student supplied the reflection themes and retains thought, planning, understanding and final quality-assurance responsibilities. The course starter supplied the original module sequence.